

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy and Safety of Remibrutinib in Patients With Secondary Progressive Multiple Sclerosis **Public Title:** A Study to Evaluate the Efficacy and Safety of Remibrutinib in Secondary Progressive Multiple Sclerosis **Short Title/Acronym:** REMASTER **Protocol Number:** CLOU064P12301 **NCT Number:** NCT07225504 **Posting ID:** 179655037608760523 **Sponsor Name:** Novartis (Novartis Pharmaceuticals) **Company:** Novartis **Investigational Product:** Remibrutinib **Phase of Development:** Phase III **Trial Status:** Not Yet Recruiting (Not yet recruiting participants) **Medical Monitor:** Novartis Clinical Operations Lead / Investigative Site Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of remibrutinib in delaying the time to 6-month confirmed disability progression (6mCDP) on the Expanded Disability Status Scale (EDSS) in patients with Secondary Progressive Multiple Sclerosis (SPMS). **Secondary Objectives:** To evaluate the safety, tolerability, and additional efficacy metrics of remibrutinib, including cognitive processing speed (SDMT), upper/lower limb function (9-HPT and T25FW), and incidence of adverse events.

2.2 Background & Rationale To address the unmet medical need for effective treatments in Secondary Progressive Multiple Sclerosis (SPMS) capable of slowing steady disability progression. Remibrutinib, a highly selective oral Bruton's tyrosine kinase (BTK) inhibitor, has demonstrated a favorable safety profile with no severe liver safety concerns in other indications, and has the potential to effectively target and reduce the immune-driven central nervous system (CNS) inflammation underlying SPMS progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Event-driven) **Control Type:** Placebo-controlled **Blinding:** Double-blind (Core Part), followed by an Open-label (Extension Part) **Randomization:** Parallel-group, 1:1 Randomization

3.2 Planned Enrollment & Duration Target Enrollment: 1,275 evaluable participants **Number of Sites:** Global multi-center

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent must be obtained prior to any assessment performed. 2. Male or female participants aged 18 to 65 years (inclusive) at Screening. 3. Diagnosis of SPMS according to the 2017 revised McDonald criteria at Screening, with an EDSS score of 3.0 to 6.0 (inclusive). 4. Documented evidence of disability progression in the 12 months before Screening, and the absence of documented clinical relapses in the 24 months before Screening and randomization.

4.2 Exclusion Criteria 1. Unwilling or unable to undergo MRI scans as per protocol (e.g., claustrophobia, absolute contraindications such as metallic implants, metallic foreign bodies, pacemaker, or defibrillator). 2. History of clinically significant central nervous system (CNS) disease (e.g., stroke, traumatic brain or spinal injury, history or presence of myelopathy) or neurological disorders which may mimic multiple sclerosis (MS). 3. Ongoing substance abuse (drug or alcohol) or any other factor (e.g., serious psychiatric condition) that may interfere with the participant's ability to cooperate and comply with the study procedures. 4. Participants with a history of confirmed Progressive Multifocal Leukoencephalopathy (PML) or neurological symptoms consistent with PML. 5. Significant bleeding risk or coagulation disorders at Screening. 6. Women of childbearing potential (WOCBP) not using highly effective methods of contraception (failure rate < 1% per year) while taking study treatment and for at least 1 week after stopping study treatment. 7. Use of exclusionary medication prior to screening/randomization as defined in the protocol.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Remibrutinib or matching Placebo administered continuously during the event-driven Core Part, transitioning to open-label remibrutinib in the Extension Part. **Route of Administration:** Oral **Duration of Treatment:** Up to approximately 5 years (Event-driven duration). **Related/Mapped Drug Reference:** ibrutinib (Trade name: Imbruvica; ATC Code: L01EL01; EMA URL: https://www.ema.europa.eu/en/documents/product-information/imbruvica-epar-product-information_en.pdf)

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptomatic management of MS and other non-exclusionary co-morbidities. **Prohibited:** Use of exclusionary disease-modifying therapies (DMTs) for MS and

medications conferring a significant bleeding risk prior to screening/randomization as defined in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline EDSS assessment, MRI qualification
Baseline	Day 0	Randomization, First Dose, Vitals, Baseline SDMT and T25FW
Interim	Q3-Q6 Months	Safety Labs, Efficacy Assessments (EDSS, T25FW, SDMT), MRI scans, AE tracking
End of Core Study	Event-Driven (~Year 5)	Final Double-Blind Assessment, Transition to Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to 6-month confirmed disability progression (6mCDP) on the Expanded Disability Status Scale (EDSS) measured from baseline up to approximately 5 years. **Measurement Method:** Standardized neurological examination combined to determine EDSS steps (ranging from 0 to 10).

7.2 Secondary & Exploratory Endpoints 1. Time to 3-month confirmed disability progression (3mCDP) on EDSS. 2. Time to 6-month confirmed disability improvement (6mCDI) on EDSS. 3. Time to 3-month worsening by at least 20% in the Timed 25-Foot Walk (T25FW). 4. 6-month worsening by at least 4 points in the Symbol Digit Modalities Test (SDMT).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of AEs, including changes in laboratory data, vital signs, electrocardiogram (ECG), and Columbia Suicide Severity Rating Scale (C-SSRS). **Serious Adverse Events (SAEs):** Routine reporting within standard 24-hour timelines; close monitoring for hepatic safety and bleeding risks inherent to BTK inhibitor class medications. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) active to review safety, tolerability, and event-driven progression thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary time-to-event outcomes. Safety Set for all participants receiving at least one dose of the study drug. **Sample Size Justification:** $N = 1,275$ powered to capture the required number of 6mCDP events to detect a statistically significant delay in

progression with remibrutinib vs. placebo. **Handling of Missing Data:** Standard time-to-event survival analysis (e.g., Cox proportional hazards models) applying censoring rules for participants lost to follow-up or completing the study prior to a confirmed progression event.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Systematic onsite and remote clinical monitoring visits to ensure subject safety, protocol adherence, and integrity of clinical assessments (e.g., EDSS evaluator certification). **Audit Trail:** Continuous eCRF data entry locking and centralized review of MRI imaging to ensure standardized radiological reads. **Criteria Requirement:** Trial must be in PHASE3 (Criteria Type: PHASE_REQUIREMENT). Source Level: 5.

11. Study Identifiers & Governance

Primary ID: CLOU064P12301 **Registry Number:** NCT07225504 **Posting ID:** 179655037608760523 **Sponsor/Lead Organization:** Novartis **Study Title (Acronym):** REMASTER **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy and Safety of Remibrutinib in Patients With Secondary Progressive Multiple Sclerosis

12. Clinical Framework (SPMS Specific)

Primary Condition / Disease Area: Secondary Progressive Multiple Sclerosis (SPMS) **Semantic Type:** Disease or Syndrome **Preferred Name / UMLS Name:** Multiple Sclerosis **Definition:** A progressive autoimmune disorder affecting the central nervous system resulting in demyelination. Patients develop physical and cognitive impairments that correspond with the affected nerve fibers. **MeSH Code:** D009103 **SNOMED ID:** 24700007 **Diagnosis Threshold:** 2017 revised McDonald criteria with baseline EDSS score of 3.0 to 6.0 **Medical Methodology:** Selective Bruton's tyrosine kinase (BTK) inhibition **Optimization Target:** Delayed progression of neurological disability

13. Study Procedures & Methodology

Management Pathway: Standard clinical assessment pathway for SPMS disease progression tracking (clinical & radiological) **Education Protocol (HCP):** Specialized certification and training for EDSS, SDMT, 9-HPT, and T25FW raters **Education Protocol (Patient):** Education on medication adherence, MS symptom tracking, and

strict contraceptive mandates **Quality Audit Mechanism:**
Centralized evaluation of MRI scans and rater reliability for
neurological outcome measures

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 60 months (5 years) for the double-blind Core Part **Onsite Visit Cadence:** Regular intervals tracking 3-month and 6-month disability confirmation periods
Remote Monitoring Frequency: Regular safety parameter checks throughout the event-driven progression timeline **Checklist Review Interval:** Continuous surveillance of concomitant medications and adverse events (C-SSRS and laboratory parameters)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	06-Apr-2026				
Clinical Operations Lead	[Name]	_____	06-Apr-2026		Lead		
Statistician	[Name]	_____	06-Apr-2026				